

Unit 1: Introduction to Structural Biology

1. What is Structural Biology?

Structural biology is a branch of molecular biology that deals with the study of the molecular structure of biological macromolecules—primarily proteins and nucleic acids—and how their structures relate to their functions.

2. What are macromolecules?

Macromolecules include proteins, DNA, RNA, and large complexes like ribosomes. They are essential for life and perform a variety of structural and functional roles.

3. Understanding Sequence-Structure-Function Relationship

The sequence of amino acids in a protein determines its 3D structure, which in turn determines its function. Changes in sequence (mutations) can alter structure and function, leading to disease.

Diagram:

Amino Acid Sequence → Protein Folding → Function

4. Screening of Target Proteins and Their Structures

This involves identifying potential protein targets (e.g., enzymes, receptors) for drugs and retrieving or predicting their 3D structures. Tools include BLAST, UniProt, PDB, and modeling software.

5. Hands-on Activities:

- Use of PDB database to retrieve structures
- Use of visualization tools like PyMOL or Chimera

Unit 2: Structural Databases and Techniques

1. Introduction to PDB (Protein Data Bank)

PDB is a repository for 3D structural data of proteins and nucleic acids obtained through X-ray crystallography, NMR, and cryo-EM.

2. Understanding 3D Structure Files

PDB files contain atomic coordinates, occupancy, temperature factors, and annotations like ligand-binding.

3. Structural Biology Techniques

- **Macromolecular Crystallography:** Determines atomic structure by analyzing diffraction patterns.
- **NMR Spectroscopy:** Determines structure in solution.
- **Cryo-EM:** Visualizes large complexes at near-atomic resolution without crystallization.

4. How to Use RCSB and Read PDB Files

- Visit [rcsb.org](https://www.rcsb.org)
- Search by protein name or PDB ID
- Use molecular viewers to analyze structures

5. Related Calculations

- Root Mean Square Deviation (RMSD)
 - Bond angles and distances
 - Surface area and volume measurements
-

Unit 3: Molecular Docking

1. Ligand-Protein Interactions

These interactions are central to drug discovery. Ligands bind to proteins at active or allosteric sites.

2. Concept of Molecular Docking

Molecular docking predicts how a ligand binds to a receptor and estimates binding affinity.

3. Small Molecule Tools

- PubChem
- ChemSpider

- SwissADME

4. Tools for Molecular Docking

- AutoDock
- AutoDock Vina
- Schrödinger Suite

5. Installation of AutoDock

Requires Python, AutoDockTools, MGLTools. Download from autodock.scripps.edu

6. Ligand and Protein Preparation

- Remove water molecules and add hydrogens
- Ligands: Minimize energy, assign Gasteiger charges

7. Grid and Docking Parameters

- **Grid:** Defines search space around active site
- **Docking:** Defines how ligands are oriented and scored

8. Case Studies

- Docking Aspirin to COX-1 enzyme
- Virtual screening for SARS-CoV-2 Mpro inhibitors

Unit 4: Molecular Dynamics Simulation

1. Introduction to MD Simulation

MD simulates the physical movements of atoms and molecules over time.

2. Why Simulations in Biology?

To observe dynamic behavior of biomolecules, predict stability, binding, and conformational changes.

3. Atomistic Models

Detailed models representing all atoms using classical mechanics.

4. Force Fields

Mathematical formulas describing interatomic interactions:

- CHARMM
- AMBER
- GROMOS

5. Steps in MD Simulation

- System setup
- Energy minimization
- Equilibration
- Production run

6. Topology and Parameter Files

- Topology defines atom types, bonds, angles
- Parameters provide energy values for those interactions

7. Periodic Boundary Conditions

Used to mimic infinite systems and reduce edge effects

8. Solvation Methods

- **Explicit:** Add water molecules
- **Implicit:** Use mathematical models for solvent effect

9. Simulation Programs

- GROMACS

- **AMBER**
- **NAMD**

10. Case Studies

- Studying protein folding
- Ligand-binding pathway
- Membrane protein dynamics